

NICOLAS MAGNE TANG, Ph.D.

Battery Systems Engineer

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PROFESSIONAL SUMMARY

Battery Systems Engineer with **7+ years** driving BMS/EMS innovation, battery state estimation (SoC/SoH/SoS), and GTR-EVE durability compliance for EVs. Ph.D. in Li-ion modeling with proven impact: **20% capacity improvement** via ohmic drop compensation, **2 major test programs (~€ 200K)** under GTR-EVE, and **€ 500K/year savings** through data-driven test optimization. Patent holder (Fr 2006555). Seeking senior battery engineering roles.

PROFESSIONAL EXPERIENCE

Jan 2024 — Present | Pilot Function System — GTR-EVE Battery

SEGULA Technologies — Ampere (Renault Group) | Paris

- Lead validation strategy for GTR-EVE battery durability regulation across **2 major test programs (~€ 200K)**, covering EV/PHEV architectures with multiple Li-ion chemistries
- Drive **€ 500K+/year savings** through optimized WLTP break-off strategies at 23°C and -7°C, reducing test cycles while improving data quality
- Develop advanced SOH estimation methodology blending empirical, algorithmic, and ML approaches — selecting the most reliable, vehicle-adapted method for production BMS
- Validate BMS estimators (SOCE/SOCR) against experimental data, closing the gap between theoretical models and real-world battery behavior
- Architect test data pipelines using MATLAB, Python, AVL Concerto, and Power BI for real-time performance tracking

Aug 2023 — Oct 2023 | Battery Algorithm Development Engineer — BMS

Freelancer — Swiss Company | Geneva

- Designed scalable, modular BMS software/hardware architecture using MBSE for stationary energy storage
- Integrated SoC/SoH/SoS estimation algorithms from sensor data into production-ready BMS firmware
- Delivered 2 comprehensive technical reports on BMS architecture, algorithm selection, and integration strategy

Nov 2019 — Apr 2023 | R&D; Engineer — Li-ion Battery Modelling (Ph.D.)

LEPMI — Grenoble INP | Grenoble

- Defined and led testing campaigns for NMC and LFP cells using ohmic drop compensation (ODC), achieving **20% improvement** in capacity extraction under high-current discharge
- Developed parametric battery model combining ODC with ECM framework, validated across 4 cell types for BMS integration in drones and e-bikes
- Published 2 scientific articles and secured **industrial patent Fr 2006555** for rapid battery state-of-charge estimation
- Performed EIS, dQ/dV, and cycling analysis on Biologic/Arbin platforms with EC-Lab

Feb 2019 — Jul 2019 | Development Engineer — Second-Life Battery Analysis

LEPMI — Grenoble INP | Grenoble

- Developed characterization protocols (OCV, CC-CV, EIS, dV/dQ) enabling second-life Li-ion battery assessment
- Built ECM-based degradation models predicting battery behavior across 500+ experimental conditions
- Conducted aging tests simulating EV second-life scenarios for battery repurposing qualification

Jul 2018 — Sep 2018 | R&D; Engineer — UAV Energy Management

Lab of Fluid Mechanics & UAV | Barcelona, Spain

- Analyzed energy consumption of fixed-wing drones with hybrid propulsion (electric + turbine)
- Developed energy system models using Simulink and MBSE for performance optimization
- Performed storage system sizing balancing power consumption, weight, and flight autonomy

Feb 2018 — Jun 2018 | Engineer — PV Energy Analysis

Centrales Villageoises | Vercors, France

- Technical-economic study for collective photovoltaic energy production (project €175K)
- Estimated PV production, analyzed energy consumption patterns, and assessed financial viability

Jun 2015 — Feb 2017 | Electronic Engineer — Instrumentation & Automation

Mi Bus UIO Start-up | Quito, Ecuador

- Designed and deployed people-counting and GPS tracking system across **120 buses** using MQTT and CAN bus
- Drove **\$50K/year savings** per operator through route optimization and data-driven scheduling
- Led full hardware lifecycle: prototyping, embedded C firmware, cloud integration, and field testing on 25 pilot buses

Mar 2014 — Mar 2015 | Engineering Intern

Escuela Politecnica Nacional | Quito, Ecuador

- NASA Space Apps Challenge 2015 — Local Winner Ecuador & Global Nominee. Project Dropi Alone: ozone leaf injury analysis via image processing

EDUCATION

Ph.D. in Energy, Process and Fluid Mechanics (BESS) — [University Grenoble Alpes](#) | 2019-2022

M.Sc. in Electrical Engineering — Smart Grids and Buildings — [ENSE3 Grenoble INP](#) | 2017-2019

B.Sc. in Electronics, Automation and Control — [National Polytechnic School](#) | 2009-2015

TECHNICAL SKILLS

Battery: BMS/EMS, SoC/SoH/SoS Estimation, GTR-EVE Compliance, Li-ion/LFP/NMC, ECM, EIS, WLTP, Cell Balancing, Aging Analysis, Second-Life

Tools: MATLAB/Simulink, Python, C/C++, AVL Concerto, Power BI, COMSOL, EC-Lab, LabVIEW, AutoCAD, PVSyst, PSIM, CANalyzer

Methods: MBSE, FMEA, Data-Driven Optimization, Sensor Integration, MQTT, IoT

Languages: Spanish (Native) | English (C1) | French (B2)

Patents: Fr 2006555 — Battery SoC Estimation (2022)